

DATA 298A — Demo Schedule and Requirements

Fall 2026. Companion to the DATA 298A introduction deck and to the project list for your section (11, 21 or 31). All dates below are the official course schedule dates.

Course schedule

Date	Day	Activity or deliverable
8/27	Thu	First online meeting — course overview, teams, topic lists released
9/3	Thu	Topic confirmed by email; Linear board created and ISA invited
9/9	Wed	Project Abstract due
9/10	Thu	Candidacy Form due
9/10	Thu	CHECKPOINT 1 — Abstract Presentation, 9:00–11:45 am
9/24	Thu	Progress Report 1 due
10/7	Wed	Workbook 1 due
10/8	Thu	CHECKPOINT 2 — Team Meeting 1, 9:00–11:45 am
10/22	Thu	Progress Report 2 due
11/11	Wed	Workbook 2 due
11/12	Thu	CHECKPOINT 3 — Team Meeting 2, 9:00–11:45 am
12/7	Mon	Final Report due
12/7	Mon	Project Presentation slides due
12/8	Tue	CHECKPOINT 4 — Project Presentation (Conference Day), 9:00 am–12:00 pm

The four shaded rows are the graded presentation sessions. Everything else is a submission deadline or a milestone with no live session.

How the checkpoints work

There are four graded presentation sessions. Each is a live demonstration of working software, not a review of slides describing working software.

- **Everyone presents.** All four members attend and each member presents the part they did. A member who does not present does not receive credit for that session.
- **Format.** Ten slides or fewer, 17 minutes of presentation and 3 minutes of questions, per audience. Team Meetings 1 and 2 have two presentations: one with the Professor and one with the ISA.
- **Submit before each session:** The GitHub repository link at the presentation commit, the Linear board link, and a one-to-two page memo.
- **Linear is the record.** Every task shown must exist in Linear as a closed issue with a named assignee and a linked artifact — a pull request, commit, dataset version or experiment run. Work that is not in Linear does not count as individual contribution.
- **Nothing is staged.** Anything you run must be runnable again on request. If part of a demo is mocked or hardcoded, say so out loud before you run it.
- **Grading.** Each rubric section is rated poor, good or excellent, per section rather than per report, as defined in the introduction deck. Poor means something required is missing or wrong; good means it meets the expectation with no major flaws; excellent means it goes beyond in a way a reader can point to. Most sections should be good.
- **Attendance.** There is no postponement. A missed session scores zero.
- **Reports.** The written report must be your own writing. ChatGPT is not permitted for writing reports.

Linear cycles

Run two-week cycles anchored to the checkpoint dates. Each cycle should close with the work below finished and linked.

Cycle	Dates	Should be finished by the end of the cycle
Cycle 1	8/27 – 9/9	Team formed, topic confirmed, repo and Linear board up, abstract written
Cycle 2	9/10 – 9/23	Data sources secured, licences checked, first extraction running
Cycle 3	9/24 – 10/7	Cleaning and EDA complete, splits fixed, metrics chosen; Workbook 1
Cycle 4	10/8 – 10/21	Transformation and cloud pipeline automated; evaluation harness runs
Cycle 5	10/22 – 11/4	Prototype v0 deployed; baseline reproduction attempted
Cycle 6	11/5 – 11/11	First model trained; prototype hardened; Workbook 2
Cycle 7	11/12 – 11/25	Benchmark table v1, multiple seeds, one ablation
Cycle 8	11/26 – 12/8	Failure study, reproducibility check, final report, 298B plan

Checkpoint 1 — Abstract Presentation, Thursday 9/10, 9:00–11:45 am

Abstract due Wednesday 9/9. Candidacy Form due Thursday 9/10.

1.1 Topic and problem statement

- State your topic number and title exactly as it appears in your section's project list, and the date it was confirmed by email.
- Restate the problem in your own words in no more than three sentences.
- Name the four models listed on your topic slide, and say which one you will build first and why.
- State numerically what success looks like: which metric, on which benchmark, at roughly what level.

1.2 The abstract

- Purpose, tasks, outcomes and applications. Ideally 200–250 words, no more than 400. APA format, following the provided template.
- Present each component — introduction, importance, goal, issues, methodology, expected outcome — on its own slide.

1.3 Initial state of the art

- Review 8 to 12 papers or production systems, one short paragraph each.
- Identify the single published baseline you will reproduce, with the exact table or figure its number comes from.
- Confirm you can actually obtain the benchmark or dataset named on your topic slide. Show that you have accessed it.

1.4 Project setup and risks

- GitHub repository with a pinned environment and a README someone outside the team can follow.
- Linear board created, ISA and instructor invited with admin access, cycles turned on, every member holding assigned issues.
- Compute budget: what hardware you have, what you will need, how you close the gap.
- The three largest risks — usually data access, compute or model size — and a specific mitigation for each.

Checkpoint 2 — Team Meeting 1, Thursday 10/8, 9:00–11:45 am

Progress Report 1 due 9/24. Workbook 1 due Wednesday 10/7. Two presentations: one with the Professor, one with the ISA.

2A Presentation with the Professor

- **Requirements.** Goals, scope and success criteria, stated numerically (for example: web application only; success is $AUC \geq 0.85$ and $p95 \text{ latency} < 120 \text{ ms}$).
- **Deliverables.** The four or more models from your topic slide: purpose, target metrics, artifacts.
- **Literature survey.** State-of-the-art papers relevant to your domain: problem, method, results, and fit to your project.
- **Related products.** Existing products and services in the space: features, gaps, and how you differentiate.
- **Data.** Datasets: size, source, schema, licence, cleaning, splits, augmentation, validation checks.
- **Organization plan.** Team roles and responsibilities, and the toolchain: GitHub for pull request reviews, Linear for cycles and issues, experiment tracking, CI/CD, containers and deployment.
- **Schedule.** Milestones, dependencies and checkpoints, mapped to the course dates.
- **Major challenges.** Data risks such as imbalance and drift, operational risks, and ethical risks such as bias and privacy — each with a mitigation and a fallback.

2B Demonstration with the ISA — data extraction, cleaning and quality

Data collection

- Identify raw data sources: APIs, databases, files, simulators, web scraping, sensor logs.
- State the licence or terms of use for each source and confirm you may use it.
- Define parameters: features, formats, timestamps, sampling method.
- Assess quantity — volume, variety and velocity. Present sample raw records.

Preprocessing

- Handle missing values, and state the imputation or deletion strategy.
- Remove duplicate or irrelevant records; report how many.
- Normalize, standardize or encode categorical variables.
- Check consistency, outliers and errors with validation tools.
- Show samples before and after.

Data quality assessment

- Exploratory analysis: distributions of key features, correlations, class imbalance and bias.
- Visualizations — histograms, boxplots, heatmaps — showing the problems you found, not only the ones you fixed.
- Any population or coverage gap in the data, and who it would disadvantage if the system were deployed.

Splits and metrics — required

- Fix train, validation and test splits now and put them under version control.
- Say how you checked for leakage between splits, and what you found.
- Choose your evaluation metrics now, before you have results, and justify each. Use the metrics named on your topic slide unless you can argue for better ones.
- Name the benchmark you will report on. If you constructed a held-out set yourself, explain why it is fair.

Checkpoint 3 — Team Meeting 2, Thursday 11/12, 9:00–11:45 am

Progress Report 2 due 10/22. Workbook 2 due Wednesday 11/11. Two presentations: one with the Professor, one with the ISA.

3A Presentation with the Professor — models and evaluation

- **Data summary.** One slide showing collection → preprocessing → transformation → splits, with key statistics: counts, class balance, leakage checks.
- **Model proposals.** Four or more models. For each: concept, features, architecture, training algorithm.
- **Model supports.** Platform, environment and tooling, with an architecture diagram showing components and data flows.
- **Comparison.** A comparison table covering targeted problems, features, approach type (statistical / ML / DL), strengths and limitations, plus a short justification of when to choose each.
- **Evaluation methods.** Planned metrics and procedures, validation strategy (cross-validation or temporal split) and the error-analysis plan.
- **First model.** Your first trained model, with learning curves, hyperparameters searched and compute consumed. It must be trained, fine-tuned or adapted by your team — a hosted API call is not one of your four models. This is the first of the four; the other three arrive in 298B.
- **Baseline reproduction.** The published number, your reproduction, the gap, and your explanation. If you could not reproduce it, present that as a finding with evidence.

3B Demonstration with the ISA — pipeline and prototype

Data transformation

- Format conversion (CSV, JSON, Parquet, Delta, TFRecord).
- Feature engineering, tokenization, augmentation.
- Aggregation and filtering, including time-series handling.
- Show samples of transformed rows or records.

Cloud pipeline

- Ingestion from source through to a warehouse or lakehouse: BigQuery, Redshift, Snowflake, S3, GCS or Delta Lake. Show file layout, tables and schemas, and verify the data is stored correctly.
- Transformation logic in DAGs or scripts, running on a schedule. A pipeline someone runs by hand is not automated.
- Monitoring and management: Airflow or Cloud Composer, with logs, status, retries and alerts. Show a real log from a real run.
- Show one sanity check failing when you deliberately corrupt an input.

Prototype

- One complete path through the system: input in, model runs, scored result out, through an interface.
- Deployed and reachable — not running on one laptop.
- It runs on output from the pipeline you just showed.
- Run it live on an input the audience chooses.

It counts if it produces a real scored output, is deployed with code in GitHub and a pinned environment, could be cloned and run by another team from your README, and contains at least one model you built. It does not count if it is a laptop notebook, a set of screenshots, an interface with hardcoded outputs, an API call with a UI on top, or something that only works on rehearsed inputs.

3C Individual contribution review

- Each member walks through their own Linear issues and the pull requests linked to them.

- The Linear history, the Git history and the demo should agree. Where they disagree, expect questions.
- If workload is unbalanced, this is the meeting to say so. It can be fixed on 11/12. It cannot be fixed on 12/8.

Checkpoint 4 — Project Presentation, Conference Day, Tuesday 12/8, 9:00 am–12:00 pm

Final Report and presentation slides due Monday 12/7. Focus on progress since Team Meeting 2; do not repeat earlier content except where needed for context.

4.1 Background and executive summary

- Briefly restate the project context and goals.
- Summarize what has been achieved, at a high level.

4.2 Model strategy, benchmark and ablation

- The models considered and the rationale for your final approach, including anything genuinely novel in it.
- The benchmark table: your model against the reproduced baseline, on the benchmark you named, using the metrics you committed to in Team Meeting 1.
- Multiple seeds with error bars, and the number of runs stated. A single lucky run is not a result.
- One ablation: remove a component and report the delta, including when nothing changes.

Use the metrics appropriate to your topic. Your topic slide is authoritative; the table below gives the usual choices.

Topic type	Report at minimum
Classification / detection	Accuracy, precision, recall, F1, confusion matrix, PR curve; per-subgroup breakdown where people are involved
Ranking / retrieval / search	Recall@k, NDCG, MRR; latency at the serving configuration you would actually deploy
Generation / LLM systems	Task-specific accuracy, plus a human or LLM judge with reported inter-rater agreement; cost per query
Robotics / control / RL	Success rate, SPL or equivalent, sample efficiency, robustness under perturbation
Systems / infrastructure	Throughput, latency percentiles, cost per unit of work, and the quality delta versus the unoptimized baseline
Security / safety	Detection rate and false-positive rate at a stated operating point, plus generalization to unseen attacks

4.3 Failure study

- At least 50 failure cases sampled from your evaluation runs.
- Categorized into a taxonomy you define, with the distribution reported.
- The top three causes named, and what you would do about each.
- The conditions under which your system should not be used.

4.4 Reproducibility

- Another team clones your repository and runs it without your help, then reports what happened.
- Pinned environment, seeds, logged experiments, documented hardware.
- Any result you cannot regenerate on request is treated as not obtained.

4.5 Where the system loses

- An explicit account of where your approach is worse than the baseline or an alternative.
- Reporting only wins is graded down. A comparison with no losses in it has usually not been run honestly.

4.6 Final prototype

- Hardened: it handles inputs you did not anticipate without crashing, and fails visibly rather than silently.
- Run live on inputs chosen by the audience.
- Teams that have already shown a fully working end-to-end cloud pipeline may show screenshots of it plus evaluation results; teams that have not must show the full live pipeline.

4.7 The 298B plan

- Which three remaining models you will build, on which benchmark, against the baseline you reproduced this semester. The model you built in 298A is the first of the four.
- The compute budget and the schedule, entered as Linear milestones before the presentation.
- The one result you most want to be able to report at the end of 298B.

What carries into 298B

298B continues the same project, the same repository and the same Linear board. Nothing restarts.

- **Linear continues.** The board stays open across the winter break and the instructor and ISA remain admins. Cycles resume in the first week of 298B.
- **Four models complete.** The three remaining models from your topic slide, built and benchmarked against the baseline you reproduced in 298A, on the same benchmark and the same fixed splits.
- **Retrain, do not just train.** Wire the pipeline so a model can be retrained on new data without hand-holding: retraining on a schedule or on drift, versioned datasets and model artifacts, an evaluation gate before promotion, and rollback when the new model is worse.
- **Full application.** The prototype becomes a complete application with a real user interface on top of the pipeline.
- **Final benchmark.** All four models measured, evaluated and compared, with the analysis and interpretation of results, an expanded failure study, and publication.

Anything you leave undone in 298A becomes 298B work on top of a heavier semester. The teams that struggle in 298B are the ones that arrived without a working prototype.

Common reasons teams lose marks

- The prototype only works on the rehearsed inputs.
- Metrics were chosen after the results were seen.
- The baseline was implemented by the team, badly, and then beaten.
- One seed, no error bars, and a small improvement claimed as a result.
- The failure study is a paragraph rather than 50 categorized cases.
- Linear was filled in the night before the presentation.
- One member did most of the work and the team hoped nobody would ask.
- The report is fluent, general, and clearly not written by the team.

None of these are difficult to avoid. All of them are easy to detect.